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QUALITY MANAGEMENT SYSTEM PROCEDURE

Title: COMPONENT LABEL REQUIREMENTS	
DOCUMENT NUMBER: 960-00099	REVISION: 06
PROCESS OWNER: MANUFACTURING TECHNICAL OPS	Page 1 of 8
APPROVERS: HARDWARE ENGINEERING, SUPPLY CHAIN, CORPORATE QUALITY	

1. **SCOPE**: This document applies to the labeling used to identify Amazon Robotics products that are purchased from suppliers, vendors and/or manufacturers. These products include, but are not limited to, components, sub-assemblies and final assemblies.

2. **PURPOSE**: This document specifies how components, sub- assemblies and assembled products are to be labeled. This is to ensure a common practice that is adhered to and understood by all affected parties. Forms of labeling other than what is described within this document can be used, as long as they convey the required information and are approved by Amazon Ro prior to use.

3. **REFERENCE DOCUMENTS**:

900-00163	KIVA PRINTED CIRCUIT BOARD DESIGN GUIDELINES
960-0015-02	SUPPLIER HANDBOOK
960-0015-05	PRODUCTION PART APPROVAL PROCESS
960-0015-08	SUPPLIER PRODUCT COMPLIANCE
960-0075	ENGINEERING DRAWING STANDARDS
960-00098	COMPONENT TRACEABILITY IDENTIFICATION FORMAT SPECIFICATION
GS1-128 (UCC/EAN-128)	SYMBOLOGY SPECIFICATION

4. **DEFINITIONS**:

   a. AR – Amazon Robotics

   b. Supplier: Any party supplying products to Amazon Robotics. Can be manufacturer, vendor or service provider.

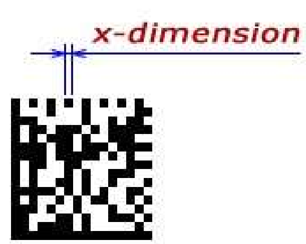
   c. Off-the-Shelf (OTS) Part: An off-the-shelf part is a part not designed by Amazon Robotics.

   d. Direct Part Marking (DPM): Displaying the content directly onto the component without the use of a label.

   e. Component Marking(s): Data defined in this document and used to mark the identity and revision of the assembly.

   f. Traceability Identifier: Data that is adhered to an assembly to represent various identifiers (vendor, date of manufacture, count etc), as defined and specified in 960-0098 and the classification of the assembly in question.

   g. X-dimension (Cell size): Width of the smallest element in a DataMatrix bar code.



   h. Quiet Zone: Quiet Zones are the space immediately around a barcode symbol that are free of printing or marks (blank space or white space).

5. **RESPONSIBILITIES**:

   a. All AR employees or contractors shall ensure required component labels meet or exceed this requirement.

   b. Supply Chain is responsible for communicating the requirements to the supplier.

   c. Production Part Approval Process will be used to evaluate labels when applicable.

   d. Supplier is responsible for complying with the requirements of this document.

6. **REQUIREMENTS**:

   a. Approval: All component labels or part markings are to be approved by AR prior to use. If a supplier is currently using a label that meets or exceeds this document, it may be used as long as it is approved by Amazon Robotics.

   b. Label or Park Marking Life Expectancy: Labels are required to be legible and adhered for a minimum of 10 years under normal operating conditions. This included both human readable and barcode formats. A white polyester with black thermal transfer with aggressive adhesive is recommended.

   c. Material: Unless otherwise specified by AR, the label material is to be defined by the supplier, however it must meet the following requirements. All labels must meet the materials compliance requirements as stated in 960-0015-08. Label material must be water, humidity, solvent and ultraviolet resistant. Glossy white polyester with aggressive adhesive is recommended.

   d. Barcode font shall be Code 128A of a size that is able to be scanned by Amazon Robotics approved scanners. Unless otherwise specified or approved, barcode shall comply with the standard Code-128A barcode specification.

   e. Human readable data: Unless otherwise specified or approved, all human readable data is to be a minimum of 6 pt font. Desired font size is 8 pt or larger. Font style to be Arial or Courier.

   f. Traceability Identifier: The data included on the traceability identifier is obtained from the tables in 960-00098, based on the part number prefix and the traceability sub-type called out on the respective assembly drawing.

   g. Component marking shall be specified on the component documentation, which may include but is not limited to the following:

      i. Part or Assembly Drawing.

      ii. Specification.

      iii. Deviation.

      iv. Purchase Order.

   h. At a minimum, component marking shall include the following:

i. Vendor number. (It is sufficient if the assembly has a Traceability Identifier that contains the Vendor number)

ii. Part Number.

iii. Revision

7. **ADHERED COMPONENT LABELS:**

a. Due to the varying sizes of components used within Amazon Robotics products, the exact size and shape of the label shall be dependent on the component itself. The label shall be placed in an easily viewable location. If the label location is specified on the component drawing, the label must be placed in that location.

b. Regardless of the label shape and size, content must be included as both barcode (primary) and human readable (secondary) font unless otherwise specified.

c. If the component requires a label or part marking for traceability identification (see 960-00098 Component Traceability Identification Format Specification), this label or part marking shall be independent of the component label. Traceability identification data to be separate from the component markings.

d. Components that require serialized traceability, the traceability label must contain the serial number in barcode form and human readable font. If space allows, it is desired to also contain the part number in human readable format only.

e. If additional label content is required, it shall be specified on the component documentation described in 7.a.

f. Examples: See Appendix.

8. **DIRECT PART MARKING (DPM):**

a. For some components, direct part marking will be preferred method of labeling. This can include but not limited to stamping, casting, printing, embossing, etching and scribing. In these cases, all applicable requirements for appearance and content shall be held. Vendor to get written approval for variances from the specified format in this document.

b. Barcoding is still the primary and should be applied whenever possible. However, with DPM it is not required in all cases.

c. Location, method and specific content shall be defined in component documentation described in 7.a.

9. **DataMatrix CODE (2D bar Code) TYPE ECC200**

a. Type ECC200: has enhanced error correction capacity to eliminate distortion and restore data if part of the code is damaged. ECC200 is internationally standardized and is generally the DataMatrix code of choice.

b. Used when capturing large amounts of manufacturing or test data within the received component.

c. Application method may vary and include yet not limited to an adhered label, direct part marking, laser engraved.

d. DataMatrix X-dimension (cell size) must be minimum of 0.03 inch unless otherwise specified and approved by AR. It’s recommended to use cell size bigger than 0.03 inch wherever possible.

e. The minimum required Quiet Zone for the Data Matrix symbol is equal to X-dimension (cell size) of the DataMatrix. So, for a 0.03” symbol, a minimum Quiet Zone of 0.03” must be maintained around the top, bottom, left, and right of the symbol

f. The specific size, shape, location, and information captured within shall all be identified on the components mechanical drawing, in the components requirements doc, or specification.

g. Examples: See Appendix.

10. **REGULATORY MARKINGS:**

a. Where applicable, any regulatory and safety markings shall be applied to the component. This includes RoHS, WEEE, Danger, Warning or similar labels.

Appendix: Adhered Component Label Example:

1: Example Serialized Traceability; Example using Part number 500-0020, it requires two labels, one containing Part-number and revision and a second containing the traceability identifier:

Part Number: 500-0020

Rev: 03

Part Number and Revision

PN: 500-0020 SN: KIV130300001

Traceability Identifier

(Note: Above traceability example contains the optional, but recommended PN).

2: Example of minimum component marking; Date Limited Direct Product Marking:

Part# 420-00123 Rev: 03

Vendor#: KIV

1310 *(traceability data here, example of 960-00098 Type 06 Date Limited, e.g. 2013 Week 10)*

3: Example of 2D barcode marking; In addition to the DataMatrix symbol, human readable text is required to include the part number, revision and traceability information. If a component or assembly is labeled or permanent marked with the human readable Amazon Robotics part number, revision or traceability information that matches the data from the 2D barcode, it is acceptable for those fields to be removed from the 2D barcode label (see example below). If needed for spacing constraints, it is also acceptable, although not recommended, for the 2D barcode and human readable text to be done as two individual labels.

SN: N17190100007

DataMatrix and AR Serial Number due to Part/Assembly marked with corresponding part number and revision separately

If the component or assembly is not labeled elsewhere, the 2D barcode human readable text must include the Amazon Robotics part number, revision and traceability information as one or two individual labels. See example below:

PN: 600-01354
Rev: 06
SN: N17190100007

DataMatrix and AR Part Number, Revision and Serial Number

REVISION HISTORY

Rev.	ECO #	Description of Change	Sections Affected	Originator/Department
01	ECO-002240	Initial Revision	All	M. Sandmann/Manufacturing Tech OPS
02	ECO-003044	Including vendor number on label	4.c, 7.b	M. Sandmann/Manufacturing Tech Ops
03	ECO-003337	Update proprietary, changed serialized label format	1, 6,7 and Appendix	P. Groudas/ Tech Ops
04	ECO-014447	Included a new section with DataMatrix requirements, example 3 in Appendix, updated document formatting	1, 4-7, 9 and Appendix	B.Harty/ Technical Operations
05	ECO-018786	Included the details about 2D Data matrix and required X-dimension, quiet zone	Definitions, 9.c, 9.d	P. Narayan, P. Muttha/APQE
06	ECO-018933	Update materials compliance requirements	6.c	K.Landry/Materials Compliance